Uncertainty in Measurement Worksheet #1

1. Determine the number of significant figures in each of the following:

a) 3427 = b) 0.00045 =

c) 172 = d) 0.982×10^{-3} =

e) 3100×10^{2} =

2. Convert each of the following into scientific notation. (Remember you sig. figs!!)

a) 3427 = b) 123 453 =

c) 172 = d) 0.502 =

e) 3100×10^{2} = f) 107.2 =

3. Convert each into decimal form.

a) 1.56×10^{4} = b) 0.00259×10^{5} =

c) 0.00259×10^{3} =

4. Calculate the following. Give the answer in correct scientific notation.

a) 4.53×10^{5}
$\underline{+2.2 \times 10^{6}}$
=
=

b) 1913.0
$\underline{+ \quad 4.6 \times 10^{3}}$
=
=

c) 2.34×10^{24}
$\underline{+1.92 \times 10^{23}}$
=
=

d) 2.130×10^{3}
$\underline{- \quad 6.6 \times 10^{2}}$
=
=

5. Calculate the following. Give the answer in the correct scientific notation.

a) $(3.95 \times 10^{2}) / (1.5 \times 10^{6})$ =

b) $(4.44 \times 10^{7}) / (2.25 \times 10^{5})$ =

c) $(3.5 \times 10^{2}) (6.45 \times 10^{10})$ =

CHEMISTRY : MEASUREMENT PRACTICE I

SIGNIFICANT DIGITS

Significant figures are the digits in any measurement that are known with certainty plus one digit that is uncertain.

Rule 1: In numbers that do not contain zeros, all the digits are significant.

3.1428 [5]
3.14 [3]
469 [3]

Rule 2: All zeros between significant digits are significant.

7.053 [4]
7053 [4]
302 [3]

Rule 3: Zeros to the left of the first nonzero digit serve only to fix the position of the decimal point and are not significant.

0.0056 [2]
0.0789 [3]
0.000001 [1]

Rule 4: In a number with digits to the right of a decimal point, zeros to the right of the last nonzero digit are significant.

43 [2]
43.0 [3]
43.00 [4]
0.00200 [3]
0.40050 [5]

Rule 5: In a number that has no decimal point, and that ends in zeros (such as 3600), the zeros at the end may or may not be significant (it is ambiguous). To avoid ambiguity express the number in scientific notation showing in the coefficient the number of significant digits.

3.6×10^3 contains two significant digits

A. How many significant digits are in each of the following numbers?

1837	
3.14145×10^4	
6005	
0.08206	
0.000014	
149356	
8.7300	
0.00743	
302400	
8.732	
14.000	
19.7342	

205.8	
1900	
1200.43	
6000	
632	
14.163000	
14.000	
302400.00	
0.0019872	
20000	
426.1	
60.0	

SCIENTIFIC NOTATION

B. Convert the following numbers into or out of scientific notation:

142.63	
1,500,000	
0.00336	
1.63×10^7	
3.11×10^{-4}	
0.00125	
86,400	
1.01×10^6	
9.81×10^1	
0.000000000000144	
4,663,310.56	

ROUNDING

GENERAL RULES FOR ROUNDING:

XY-------> X
When Y > 5, increase X by 1
When Y < 5, don't change X

When Y = 5,

If X is odd, increase X by 1
If X is even, don't change X

C. Round each of the following numbers to four significant digits.

6.16782	
6.19648	
0.0019872	
3.14145×10^4	
213.25	
14.163000	
90210	
234.4	
1200.43	
0.0022475	
14.16300	
0.02315	
13.462	
135.69	
152.00	
395.55	

CHEMISTRY : MEASUREMENT PRACTICE III

A. Determine the number of significant digits in the following numbers:

__________ 1) 5600

__________ 2) 45.00

__________ 3) 105.0

__________ 4) 0.00565

__________ 5) 0.005400

__________ 6) 89.543

__________ 7) 5, 056, 300

__________ 8) 95.0540

__________ 9) 93,000,000

B. Perform the indicated operations and express your answer to the correct number of significant digits:

__________ 10) (6.92)(7.9)

__________ 11) (8.245)/(9.00)

__________ 12) (4.46)/(6.52)

__________ 13) (9.825)/(8.20)

__________ 14) (8.95) (9.162)/(4.25) (6.3)

C. Perform the indicated operations and express your answer to the correct number of significant digits:

__________ 15) 5.50 + 0.528 + 9.2

__________ 16) 420 + 8900 + 620

__________ 17) 0.00526 - 0.52

__________ 18) 820.0 + 19.5 + 6

__________ 19) 4,285.75 - 520.1 - 386.255

__________ 20) (0.526) (895) + 20.8

__________ 21) 3.414 s +10.02 s + 58.325 s + 0.000 98 s

__________ 22) 1884 kg + 0.94 kg +1.0 kg + 9.778 kg

__________ 23) 2104.1 m - 463.09 m

__________ 24) 2.326 h - 0.104 08 h

__________ 25) 10.19 m x 0.013 m

__________ 26) 140.01 cm x 26.042 cm x 0.0159 cm

__________ 27) 80.23 m ÷ 2.4 s

__________ 28) 4.301 kg ÷ 1.9 cm^3

__________ 29) 3.68 + 7.3645 + 0.5

__________ 30) 0.243 + 76.720 + 4.6494

__________ 31) 14.745 - 1.60

__________ 32) 0.5642 - 0.260

__________ 33) 6.02 x 2.0

__________ 34) 0.65 x 427

__________ 35) 0.022 x 0.467

__________ 36) 174 ÷ 24

__________ 37) 420 ÷ 17.5

__________ 38) 3.0899 mm x 22.4 mm

__________ 39) 3.4500 cm^2 ÷ 450 cm

__________ 40. 13.80 cm - 6.0741 cm

D. For each item below determine the number of significant digits in the number or answer to the problem: (don't do the calculation)

__________ 41) 804.58

__________ 42) 250.00

__________ 43) 3000

__________ 44) 10.00 m x 84.767 m

__________ 45) 0.00300900870

__________ 46) 400 x 87.488

__________ 47) 160.0001

__________ 48) 3.0 x 4.000

__________ 49) 0.00560

__________ 50) 0.7600 ÷ 1.50